

ARITHMETIC CORE — QUESTION BANK (120 solved questions)

Percentage · Profit–Loss–Discount · Ratio–Proportion–Partnership ·

Averages–Ages · Mixtures–Alligation–Variation.

Every numeric answer computed and verified with Python before printing.

Q1. 40% of a number is 96 more than 25% of the same number. Find the number.

A. 560 B. 384 C. 736 D. 640

Answer: D — 640

Trick: The gap between two percentages of the SAME number is itself a percentage of that number: $40 - 25 = 15\%$ = 96, so $1\% = 6.4$.

Solution: $15\% \text{ of } N = 96 \rightarrow 1\% \text{ of } N = 6.4 \rightarrow N = 640$.

Trap: 736 — finding 640 first and then adding the 96 difference on top instead of solving $15\% \text{ of } N = 96$.

Q2. A number is increased by 40% and the result is then decreased by 40%. If the final value is 336, find the original number.

A. 384 B. 400 C. 420 D. 480

Answer: B — 400

Trick: $+a\%$ then $-a\%$ always leaves $(100 - a^2/100)\% = 84\%$ here — divide the final value by 0.84.

Solution: Net factor = $1.4 \times 0.6 = 0.84 \rightarrow N \times 0.84 = 336 \rightarrow N = 400$.

Trap: 420 — dividing by 0.8, i.e. undoing only the 40% cut and ignoring the earlier rise.

Q3. The price of sugar rises by 25%. By what percentage must a family cut its consumption so that the expenditure on sugar stays the same?

A. 16.67% B. 20% C. 25% D. 22%

Answer: B — 20%

Trick: Required cut = $\text{rise}/(100 + \text{rise}) \times 100 = 25/125 = 1/5 = 20\%$ — the reciprocal of the multiplier, never the same %.

Solution: Cut = $25/(100+25) \times 100 = 20\%$.

Trap: 25% — cutting consumption by the same % as the price rise (equal-base error).

Q4. A salary is first increased by 10% and then decreased by 10%. What is the net change in the salary?

A. 2% decrease B. 1% increase C. No change D. 1% decrease

Answer: D — 1% decrease

Trick: $+a\%$ then $-a\%$ (either order) gives a sure loss of $a^2/100 = 1\%$ — the base shifts after the first change.

Solution: Net factor = $1.1 \times 0.9 = 0.99 \rightarrow 1\% \text{ decrease}$.

Trap: 'No change' — cancelling $+10\%$ and -10% as if they acted on the same base.

Q5. A's salary is 25% more than B's. B's salary is what per cent less than A's?

A. 16.67% B. 20% C. 24% D. 25%

Answer: B — 20%

Trick: Reverse of ' $+x\%$ ' is $x/(100+x)$, never x itself: $25/125 = 20\%$.

Solution: Let $B = 100 \rightarrow A = 125 \rightarrow B$ is $25/125 = 20\%$ less than A .

Trap: 25% — using the same base (B) for both directions.

Q6. In an examination, a candidate scoring 30% fails by 45 marks while another scoring 45% gets 15 marks more than the pass marks. Find the maximum marks.

A. 450 B. 300 C. 500 D. 400

Answer: D — 400

Trick: The two scores bracket the pass mark, so the % gap absorbs BOTH gaps: $15\% \text{ of max} = 45 + 15$.

Solution: $15\% \text{ of max} = 60 \rightarrow \text{max} = 400$ (pass mark = 165).

Trap: 300 — taking $15\% \text{ of max} = 45$ only and forgetting the 15-mark surplus.

Q7. The population of a town is 8,000. It increases by 10% in the first year and decreases by 8% in the second year. What is the population at the end of the second year?

A. 7,920 B. 8,080 C. 8,096 D. 8,000

Answer: C — 8,096

Trick: Chain the multipliers, never add percentages: $8,000 \times 1.1 \times 0.92$.

Solution: $8,000 \times 1.1 = 8,800$; $8,800 \times 0.92 = 8,096$.

Trap: 8,080 — adding the percentages ($+10 - 8 = +2\%$ of 8,000).

Q8. In an election between two candidates, 20% of the 7,500 votes polled were invalid. The winner got 55% of the valid votes. How many valid votes did the loser get?

A. 2,700 B. 2,500 C. 3,000 D. 3,375

Answer: A — 2,700

Trick: Fix the valid base first (80% of 7,500 = 6,000); the loser has 45% of VALID votes, not of total.

Solution: Valid = 6,000 → loser = 45% × 6,000 = 2,700.

Trap: 3,375 — applying 45% to all 7,500 votes and ignoring the invalid ones.

Q9. A man spends 75% of his income. His income rises by 20% and his expenditure by 10%. By what per cent do his savings increase?

A. 25% B. 12.5% C. 50% D. 30%

Answer: C — 50%

Trick: Take income = 100 and track each block; savings move on THEIR OWN old base ($\Delta S \div 25$), not on the new income.

Solution: I = 100 → E = 75, S = 25. New E = 82.5, new S = 37.5 → rise = $12.5/25 = 50\%$.

Trap: 12.5% — quoting the absolute rise (₹12.5 on a ₹100 base) as the percentage.

Q10. Two successive discounts of 20% and 10% are equivalent to a single discount of:

A. 2% B. 27% C. 30% D. 28%

Answer: D — 28%

Trick: Single % = $a + b - ab/100 = 20 + 10 - 2 = 28\%$.

Solution: $20 \times 10/100 - 2 = 28\%$.

Trap: 30% — simply adding the two discounts.

Q11. If 35% of a number is 182, what is 60% of the same number?

A. 336 B. 280 C. 109.2 D. 312

Answer: D — 312

Trick: Scale the part directly: $60\% = (60/35) \times 182 = 312$. Never rebuild the whole number.

Solution: $60\% = 182 \times 12/7 = 312$.

Trap: 109.2 — taking 60% of 182 (the given part) instead of 60% of the number.

Q12. 25% of a number is 8 less than $33\frac{1}{3}\%$ of 72. Find the number.

A. 72 B. 88 C. 64 D. 96

Answer: C — 64

Trick: $33\frac{1}{3}\% = 1/3 \rightarrow 24$; then 25% ($1/4$) of N = $24 - 8 = 16 \rightarrow N = 64$. Fraction forms beat decimals.

Solution: $1/3$ of 72 = 24 → $1/4$ of N = 16 → N = 64.

Trap: 96 — dividing 24 by $1/4$ and forgetting the '8 less'.

Q13. A man spends 40% of his income on food and 20% of the remainder on rent. If he saves ₹9,600, what is his income?

A. ₹24,000 B. ₹16,000 C. ₹19,200 D. ₹20,000

Answer: D — ₹20,000

Trick: Each percentage eats into what is LEFT: $0.6 \times 0.8 = 48\%$ saved → income = savings ÷ 0.48.

Solution: Left after food 60%, after rent 80% of that = 48%. $0.48 I = 9,600 \rightarrow I = ₹20,000$.

Trap: ₹16,000 — taking rent as 20% of the full income ($60\% - 20\% = 40\%$ left).

Q14. A bacteria culture of 2,500 grows by 20% every hour. What is the count after 2 hours?

A. 3,000 B. 3,500 C. 3,400 D. 3,600

Answer: D — 3,600

Trick: Growth chains multiply: $2,500 \times 1.2 \times 1.2$. The simple-interest style $2,500 + 500 + 500$ is the trap.

Solution: $2,500 \times 1.44 = 3,600$.

Trap: 3,500 — adding a flat 500 each hour instead of compounding.

Q15. A's income is 60% of B's, and B's income is 40% more than C's. A's income is what per cent of C's?

A. 100% B. 76% C. 60% D. 84%

Answer: D — 84%

Trick: Chain through one unit: $A = 0.6 \times 1.4 \times C$ — percentages multiply, they never add across different bases.

Solution: $A = 0.6 \times 1.4 C = 0.84 C \rightarrow 84\%$.

Trap: 100% — adding 60% and 40% as if they shared a base.

Q16. When a number is decreased by 30%, it becomes 497. The number is:

A. 570 B. 700 C. 710 D. 646.1

Answer: C — 710

Trick: 70% of N = 497 → $N = 497 \times 10/7$. Undo with the REMAINING %, not by adding 30% back.

Solution: $N = 497/0.7 = 710$.

Trap: 646.1 — adding 30% of 497 back on (wrong base).

Q17. The difference between 55% and 38% of a number is 204. Find the number.

A. 1,200 B. 1,300 C. 1,100 D. 1,360

Answer: A — 1,200

Trick: A % gap of the same number: $55 - 38 = 17\% = 204 \rightarrow 1\% = 12 \rightarrow N = 1,200$.

Solution: $17\% \text{ of } N = 204 \rightarrow N = 1,200$.

Trap: 1,360 — using a 15% gap (miscomputing $55 - 38$).

Q18. A reduction of 20% in the price of sugar enables a person to buy 5 kg more for ₹600. What is the reduced price per kg?

A. ₹36 B. ₹30 C. ₹20 D. ₹24

Answer: D — ₹24

Trick: Money saved = 20% of 600 = ₹120, and that ₹120 buys the extra 5 kg at the NEW price: $120/5 = ₹24$.

Solution: Saving ₹120 $\rightarrow$ reduced price = $120/5 = ₹24$ (original ₹30).

Trap: ₹30 — quoting the original price instead of the reduced one.

Q19. A candidate scores $33\frac{1}{3}\%$ and fails by 40 marks. Another scores 50% and gets 35 marks more than the pass marks. Find the maximum marks.

A. 450 B. 420 C. 400 D. 500

Answer: A — 450

Trick: Bracket the pass mark: $(50 - 33\frac{1}{3})\% = 16\frac{2}{3}\% = \frac{1}{6} \text{ of max} = 75 \text{ marks} \rightarrow \text{max} = 450$.

Solution: $16\frac{2}{3}\% \text{ of max} = 75 \rightarrow \text{max} = 450$ (pass = 190).

Trap: 500 — treating the 75-mark gap as 15% of the maximum.

Q20. Fresh fruit contains 80% water. How much dry fruit (containing 20% water) can be obtained from 100 kg of fresh fruit?

A. 25 kg B. 20 kg C. 80 kg D. 50 kg

Answer: A — 25 kg

Trick: Solids never change: 20 kg of solid = 80% of the dry fruit $\rightarrow$ dry = 25 kg.

Solution: Solid = 20 kg $\rightarrow$ dry fruit = $20/0.8 = 25$ kg.

Trap: 20 kg — equating the solid content with the dry-fruit weight.

Q21. A man spends 30% of his income on rent and 40% of the remainder on food. If he saves ₹6,300, find his income.

A. ₹12,600 B. ₹21,000 C. ₹18,000 D. ₹15,000

Answer: D — ₹15,000

Trick: Successive cuts: $0.7 \times 0.6 = 42\% \text{ saved} \rightarrow \text{income} = 6,300 \div 0.42$.

Solution: $42\% \text{ of } I = 6,300 \rightarrow I = ₹15,000$.

Trap: ₹21,000 — taking food as 40% of the FULL income (70% spent, 30% left).

Q22. The population of a village first decreases by 5% and then increases by 5%. What is the net percentage change?

A. 1% decrease B. No change C. 0.25% increase D. 0.25% decrease

Answer: D — 0.25% decrease

Trick: $\pm a\% \text{ in succession} = -a^2/100 = -0.25\%$ — the dip survives whichever order you apply.

Solution: $0.95 \times 1.05 = 0.9975 \rightarrow 0.25\% \text{ decrease}$.

Trap: 'No change' — cancelling -5% and +5% on shifted bases.

Q23. In a school, 30% of the students got first division and 45% got second division. If the remaining 45 students got third division, how many students appeared?

A. 180 B. 150 C. 225 D. 200

Answer: A — 180

Trick: Third division = $100 - 30 - 45 = 25\% = 45 \rightarrow \text{total} = 45 \times 4 = 180$.

Solution: $25\% = 45 \rightarrow \text{total} = 180$.

Trap: 150 — treating the 45 students as 30% of the total.

Q24. A's salary is 20% less than B's. By what per cent is B's salary more than A's?

A. 24% B. 25% C. 16.67% D. 20%

Answer: B — 25%

Trick: Reverse of '-x%' is $x/(100 - x)$: $20/80 = 25\%$.

Solution: A = 80 when B = 100 $\rightarrow$ B is 25% more.

Trap: 20% — same-base error in the reverse direction.

Q25. The price of an article is successively increased by 10%, 20% and 25%. What is the overall percentage increase?

A. 62% B. 68.75% C. 65% D. 55%

Answer: C — 65%

Trick: Multiply the factors: $1.1 \times 1.2 \times 1.25 = 1.65 \rightarrow +65\%$. Three rises never add.

Solution: $1.1 \times 1.2 = 1.32$; $\times 1.25 = 1.65 \rightarrow 65\%$.

Trap: 55% — adding $10 + 20 + 25$.

Q26. A trader sells two articles for ₹960 each — one at a 20% profit and the other at a 20% loss. What is his overall result?

A. 4% gain B. 2% loss C. No profit, no loss D. 4% loss

Answer: D — 4% loss

Trick: Equal SPs with $+x\%$ and $-x\%$ give a sure loss of $x^2/100 = 4\%$ — the two cost prices (₹800, ₹1,200) differ.

Solution: CPs: $960/1.2 = 800$, $960/0.8 = 1,200 \rightarrow$ CP 2,000 vs SP 1,920 $\rightarrow 4\%$ loss.

Trap: 'No profit, no loss' — cancelling $+20\%$ and -20% which act on different bases.

Q27. A man sells two horses for ₹1,800 each, gaining 25% on one and losing 25% on the other. Find his overall loss per cent.

A. 6.25% gain B. No profit, no loss C. 4% loss D. 6.25% loss

Answer: D — 6.25% loss

Trick: Same rule, new x : loss $= x^2/100 = 625/100 = 6.25\%$. With fractions: $(1/4)^2 = 1/16$ of total CP.

Solution: CPs $1,440 + 2,400 = 3,840$; SP 3,600 $\rightarrow$ loss ₹240 = 6.25%.

Trap: 4% — carrying the 20% template ($20^2/100$) onto a 25% question.

Q28. The marked price of an article is 40% above its cost price. If it is sold at a discount of 15%, what is the profit per cent?

A. 17% B. 19% C. 25% D. 21%

Answer: B — 19%

Trick: Take CP = 100: MP = 140, SP = $140 \times 0.85 = 119$ — the discount acts on MP, not on CP.

Solution: SP = 119 on CP 100 $\rightarrow 19\%$ profit.

Trap: 25% — subtracting the discount from the markup ($40 - 15$).

Q29. A shopkeeper marks his goods 25% above cost price and allows a discount of 16%. Find his profit per cent.

A. 6% B. 9% C. 4% D. 5%

Answer: D — 5%

Trick: CP = 100 $\rightarrow$ MP = 125 $\rightarrow$ SP = $125 \times 0.84 = 105 \rightarrow$ profit 5%.

Solution: CP 100 $\rightarrow$ MP 125 $\rightarrow$ SP $125 \times 0.84 = 105 \rightarrow 5\%$.

Trap: 9% — computing $25 - 16$ and ignoring that the discount applies to MP.

Q30. A dishonest dealer professes to sell goods at cost price but uses a weight of 800 g for a kilogram. Find his gain per cent.

A. 20% gain B. 27% gain C. 22% gain D. 25% gain

Answer: D — 25% gain

Trick: Gain% = error $\div$ FALSE weight = $200/800$ — divide by what the customer actually receives.

Solution: $200/800 = 25\%$.

Trap: 20% — dividing the 200 g error by the true 1,000 g.

Q31. A shopkeeper sells at cost price but gives only 900 g instead of 1 kg. What is his gain per cent?

A. 10% gain B. 12.5% gain C. 9% gain D. 11⅓% gain

Answer: D — 11⅓% gain

Trick: $100/900 = 1/9 = 11\frac{1}{3}\%$ — the short 100 g is earned on every 900 g delivered.

Solution: $100/900 = 11\frac{1}{3}\% \approx 11.11\%$.

Trap: 10% — dividing by 1,000 g (true weight) instead of 900 g.

Q32. A milkman buys milk at ₹40 per litre, adds water, and sells the mixture at ₹40 per litre, making a 25% profit. How much water is there in a litre of the mixture?

A. 150 mL B. 250 mL C. 400 mL D. 200 mL

Answer: D — 200 mL

Trick: Selling at cost with 25% gain means the mixture is 4/5 milk: water = 1/5 of a litre = 200 mL.

Solution: Cost of 1 L mixture = $40/1.25 = ₹32 \rightarrow$ milk 0.8 L + water 0.2 L = 200 mL.

Trap: 250 mL — taking 25% of a litre as the water added.

Q33. A 60-litre mixture contains milk and water in the ratio 7 : 3. How much water must be added to make the ratio 7 : 5?

A. 12 L B. 10 L C. 6 L D. 18 L

Answer: A — 12 L

Trick: Milk (42 L) is fixed: water must become $\frac{5}{7}$ of 42 = 30 L $\rightarrow$ add 30 – 18 = 12 L.

Solution: Water needed = 30 – 18 = 12 L.

Trap: 6 L — treating one ratio unit (6 L) as the answer.

Q34. The marked price of a cooler is ₹500. A trader allows two successive discounts of 20% and 10%. What is the selling price?

A. ₹360 B. ₹370 C. ₹350 D. ₹400

Answer: A — ₹360

Trick: $SP = 500 \times 0.8 \times 0.9 = 360$; the equivalent single discount is 28%, not 30%.

Solution: $500 \times 0.8 = 400$; $400 \times 0.9 = ₹360$.

Trap: ₹350 — applying a flat 30% (20 + 10).

Q35. If the cost price of 18 articles equals the selling price of 15 articles, find the profit per cent.

A. 20% B. 15% C. 16.67% D. 25%

Answer: A — 20%

Trick: $SP/CP = 18/15 = 6/5 \rightarrow 20\%$. The BIGGER count sits on top.

Solution: $SP/CP = 18/15 = 6/5 \rightarrow 20\%$.

Trap: 16.67% — inverting the ratio to 15/18.

Q36. By selling 33 metres of cloth, a trader gains the selling price of 11 metres. Find his profit per cent.

A. 40% B. 50% C. 25% D. 33.33%

Answer: B — 50%

Trick: Gain = SP of 11 m $\rightarrow$ CP of 33 = SP of 22 $\rightarrow$ profit = $11/22 = 50\%$.

Solution: $CP(33) = SP(22) \rightarrow$ profit = $11/22$ of CP = 50%.

Trap: 33.33% — dividing the gain by SP (11/33) instead of by CP.

Q37. Two successive discounts of 25% and 20% are equivalent to a single discount of:

A. 38% B. 40% C. 42% D. 45%

Answer: B — 40%

Trick: $a + b - ab/100 = 25 + 20 - 5 = 40\%$.

Solution: $25 + 20 - (25 \times 20)/100 = 40\%$.

Trap: 45% — adding the two discounts.

Q38. A man sells an article at 10% profit. Had he sold it for ₹36 more, he would have gained 25%. Find the cost price.

A. ₹300 B. ₹240 C. ₹144 D. ₹216

Answer: B — ₹240

Trick: The ₹36 spans the % gap: 15% of CP = 36 $\rightarrow$ CP = 240.

Solution: $CP = 36/0.15 = ₹240$.

Trap: ₹144 — dividing 36 by 25% instead of by the 15% gap.

Q39. An article is sold at a loss of 10%. Had it been sold for ₹90 more, there would have been a profit of 8%. Find the cost price.

A. ₹900 B. ₹1,125 C. ₹750 D. ₹500

Answer: D — ₹500

Trick: The ₹90 bridges –10% to +8%: 18% of CP = 90 $\rightarrow$ CP = 500.

Solution: $CP = 90/0.18 = ₹500$.

Trap: ₹900 — dividing 90 by the 10% loss alone.

Q40. A shopkeeper allows a 12% discount on the marked price and still makes a 10% profit. What is the ratio of marked price to cost price?

A. 6 : 5 B. 5 : 4 C. 4 : 5 D. 9 : 8

Answer: B — 5 : 4

Trick: $MP \times 0.88 = CP \times 1.1 \rightarrow MP/CP = 110/88 = 5/4$.

Solution: $MP \times 0.88 = CP \times 1.1 \rightarrow MP : CP = 110 : 88 = 5 : 4$.

Trap: 4 : 5 — inverting the ratio.

Q41. A trader sells 40% of his goods at a 10% profit and the remaining 60% at a 5% loss. What is his net profit or loss per cent?

A. 2.5% profit B. 1% profit C. 3% profit D. 5% loss

Answer: B — 1% profit

Trick: Weight the parts: $0.4(+10) + 0.6(-5) = +4 - 3 = +1\%$.

Solution: Net = +1% profit.

Trap: 5% loss — subtracting 5 from 10 while ignoring the quantity weights.

Q42. A fan bought for ₹1,500 needs ₹300 of repairs. It is then sold for ₹2,070. Find the profit per cent.

A. 38% B. 15% C. 20% D. 12%

Answer: B — 15%

Trick: True CP = 1,500 + 300 = 1,800; profit 270/1,800 = 15%.

Solution: CP = 1,500 + 300 = 1,800; (2,070 – 1,800)/1,800 = 15%.

Trap: 38% — dividing the profit by ₹1,500 and ignoring the repair cost.

Q43. Two successive discounts on an article total 40%. If the first discount is 25%, what is the second discount?

A. 18% B. 15% C. 20% D. 25%

Answer: C — 20%

Trick: Remaining factors multiply: $0.75 \times (1 - d) = 0.60 \rightarrow d = 20\%$.

Solution: Second discount = 20%.

Trap: 15% — taking 40 – 25.

Q44. A dealer sells goods at 20% below cost price but uses a weight of only 750 g per kg. Find his true gain or loss per cent.

A. 10% gain B. $6\frac{2}{3}\%$ gain C. 5% loss D. 6.67% loss

Answer: B — $6\frac{2}{3}\%$ gain

Trick: Price cut and short weight MULTIPLY: per true kg he collects $0.8 \text{ CP} \div 0.75 = 16/15 \text{ CP} \rightarrow 6\frac{2}{3}\%$ gain.

Solution: $0.8/0.75 = 16/15 \rightarrow 6\frac{2}{3}\%$ gain.

Trap: 6.67% loss — a sign slip from reading '20% below cost' as the dominant effect.

Q45. The profit earned on selling an article is one-fifth of its selling price. Find the profit per cent.

A. 30% B. 25% C. 20% D. 16.67%

Answer: B — 25%

Trick: Profit = SP/5 $\rightarrow$ CP = 4SP/5 $\rightarrow$ profit/CP = $(1/5)/(4/5) = 25\%$.

Solution: P = SP/5 $\rightarrow$ CP = 4SP/5 $\rightarrow$ P/CP = $(1/5)/(4/5) = 25\%$.

Trap: 20% — dividing the profit by SP instead of by CP.

Q46. By selling an article for ₹1,800 a man gains as much as he loses by selling it for ₹1,200. Find the cost price.

A. ₹1,600 B. ₹1,200 C. ₹1,440 D. ₹1,500

Answer: D — ₹1,500

Trick: Equal gain and loss $\rightarrow$ CP is the exact midpoint of the two selling prices.

Solution: CP = $(1,800 + 1,200)/2 = ₹1,500$.

Trap: ₹1,200 — taking the lower SP as the cost price.

Q47. A shopkeeper allows a 25% discount on the marked price and still makes a profit of 50%. By what per cent is the marked price above the cost price?

A. 100% B. 80% C. 75% D. 120%

Answer: A — 100%

Trick: $\text{MP} \times 0.75 = \text{CP} \times 1.5 \rightarrow \text{MP} = 2 \text{ CP} \rightarrow 100\%$ above cost.

Solution: $\text{MP} \times 0.75 = \text{CP} \times 1.5 \rightarrow \text{MP} = 2 \text{ CP} \rightarrow 100\%$.

Trap: 75% — adding the discount and profit percentages.

Q48. An article costing ₹800 is marked 60% above cost and then sold at a 25% discount. Find the profit per cent.

A. 35% profit B. 15% profit C. 25% profit D. 20% profit

Answer: D — 20% profit

Trick: SP = $800 \times 1.6 \times 0.75 = 960 \rightarrow$ profit ₹160 = 20%.

Solution: 20% profit (₹160).

Trap: 35% — subtracting the 25% discount from the 60% markup.

Q49. After allowing a discount of 10% a trader makes a profit of 8%. If the selling price is ₹1,080, find the marked price.

A. ₹1,160 B. ₹1,250 C. ₹1,200 D. ₹1,296

Answer: C — ₹1,200

Trick: MP = SP $\div$ 0.9 — once SP is known, the profit % is extra information.

Solution: MP = $1,080/0.9 = ₹1,200$.

Trap: ₹1,296 — multiplying 1,080 by 1.2 and reversing the chain wrongly.

Q50. A trader buys goods worth ₹1,200. He sells one-fourth of them at a 20% loss. At what profit per cent must he sell the remainder to gain 10% on the whole?

A. 15% B. 20% C. 18% D. 25%

Answer: B — 20%

Trick: Whole gain needed = ₹120; the quarter already lost ₹60 $\rightarrow$ the rest (CP ₹900) must gain ₹180 = 20%.

Solution: Gain needed ₹120; quarter lost ₹60 $\rightarrow$ rest must gain $180/900 = 20\%$.

Trap: 15% — averaging the two percentages $((20 + 10)/2)$.

Q51. ₹1,750 is divided among A, B and C such that $A : B = 2 : 3$ and $B : C = 4 : 5$. Find C's share.

A. ₹750 B. ₹400 C. ₹600 D. ₹900

Answer: A — ₹750

Trick: Make B common: $A : B : C = 8 : 12 : 15 = 35$ parts; unit = $1,750/35 = 50$.

Solution: $C = 15 \times 50 = ₹750$.

Trap: ₹600 — reporting B's share (12 units).

Q52. Find the mean proportional between 12 and 3.

A. 7.5 B. 4 C. 6 D. 36

Answer: C — 6

Trick: Mean proportional = $\sqrt{(\text{product})} = \sqrt{36} = 6$ — a GP middle term, not the average.

Solution: $\sqrt{(12 \times 3)} = \sqrt{36} = 6$.

Trap: 7.5 — taking the arithmetic mean $(12 + 3)/2$.

Q53. Find the third proportional to 9 and 12.

A. 14 B. 15 C. 16 D. 18

Answer: C — 16

Trick: Third proportional x satisfies $9 : 12 = 12 : x \rightarrow x = 144/9 = 16$ (GP, not AP).

Solution: $9 : 12 :: 12 : x \rightarrow x = 144/9 = 16$.

Trap: 15 — continuing the arithmetic pattern (9, 12, 15).

Q54. Find the fourth proportional to 4, 9 and 12.

A. 27 B. 3 C. 24 D. 21

Answer: A — 27

Trick: $4 : 9 = 12 : x \rightarrow x = 9 \times 12 \div 4 = 27$.

Solution: $4 : 9 :: 12 : x \rightarrow x = 9 \times 12/4 = 27$.

Trap: 3 — inverting the set-up to $4 \times 9 \div 12$.

Q55. A and B invest ₹5,000 and ₹7,000 in a business. Out of a profit of ₹2,880, what does A receive?

A. ₹1,440 B. ₹1,000 C. ₹1,200 D. ₹1,680

Answer: C — ₹1,200

Trick: Equal time $\rightarrow$ profit follows capital $5 : 7$; A gets $5/12$ of ₹2,880.

Solution: Units $5 + 7 = 12$; $A = 5/12 \times 2,880 = ₹1,200$.

Trap: ₹1,440 — splitting the profit equally.

Q56. A invests ₹10,000 for 6 months and B invests ₹8,000 for 9 months in a business. Find the ratio of their profits.

A. $5 : 4$ B. $5 : 6$ C. $6 : 5$ D. $4 : 5$

Answer: B — $5 : 6$

Trick: Profit $\propto$ capital $\times$ months: $60,000 : 72,000 = 5 : 6$.

Solution: $60,000 : 72,000 = 5 : 6$.

Trap: $6 : 5$ — inverting the ratio.

Q57. A, B and C invest in the ratio $2 : 3 : 5$. After 4 months, A withdraws half his capital. Find the ratio of their profits at the end of the year.

A. $4 : 9 : 10$ B. $4 : 9 : 15$ C. $5 : 9 : 15$ D. $2 : 3 : 5$

Answer: B — $4 : 9 : 15$

Trick: Split A's capital-time: $2 \times 4 + 1 \times 8 = 16$ against B's 36 and C's 60 $\rightarrow 4 : 9 : 15$.

Solution: $16 : 36 : 60 = 4 : 9 : 15$.

Trap: $2 : 3 : 5$ — ignoring the mid-year withdrawal.

Q58. A sum of money is divided among A, B and C in the ratio $1/2 : 1/3 : 1/4$. If A receives ₹900 more than C, find the total sum.

A. ₹3,300 B. ₹3,900 C. ₹4,200 D. ₹2,700

Answer: B — ₹3,900

Trick: Clear the fractions ($\times 12$): $6 : 4 : 3$. $A - C = 3$ parts = ₹900 $\rightarrow$ part = 300 $\rightarrow$ total = 13 parts.

Solution: Total = $13 \times 300 = ₹3,900$.

Trap: ₹2,700 — multiplying the ₹900 difference by 3 instead of scaling one part to 300.

Q59. Two numbers are in the ratio $3 : 5$. If 10 is added to each, the ratio becomes $5 : 7$. Find the numbers.

A. 6 and 10 B. 12 and 20 C. 15 and 25 D. 20 and 30

Answer: C — 15 and 25

Trick: Let them be $3x, 5x$: $(3x+10)/(5x+10) = 5/7 \rightarrow 21x + 70 = 25x + 50 \rightarrow x = 5$.

Solution: Numbers are 15 and 25.

Trap: 6 and 10 — taking $x = 2$ after a sign slip in the cross-multiplication.

Q60. The incomes of A and B are in the ratio 5 : 4 and their expenditures in the ratio 3 : 2. If each saves ₹6,000, find A's income.

A. ₹18,000 B. ₹12,000 C. ₹9,000 D. ₹15,000

Answer: D — ₹15,000

Trick: Subtract the two savings equations: $(5x - 3y) - (4x - 2y) = 0 \rightarrow x = y = 3,000 \rightarrow A = 15,000$.

Solution: $A = ₹15,000$ ($B = ₹12,000$).

Trap: ₹12,000 — reporting B's income.

Q61. If $a : b = 3 : 4$, find the value of $(3a + 2b) : (5a - b)$.

A. 17 : 13 B. 11 : 17 C. 17 : 11 D. 19 : 11

Answer: C — 17 : 11

Trick: Substitute $a = 3$, $b = 4$ straight into the expression: 17 : 11.

Solution: $3(3) + 2(4) = 17$; $5(3) - 4 = 11 \rightarrow 17 : 11$.

Trap: 11 : 17 — reading the resulting ratio backwards.

Q62. A starts a business with ₹6,000. After 5 months B joins with ₹9,000. If the year-end profit is ₹6,900, find B's share.

A. ₹2,760 B. ₹3,680 C. ₹3,220 D. ₹3,450

Answer: C — ₹3,220

Trick: Capital $\times$ months: A 72,000 : B 63,000 = 8 : 7 (B counts only 7 months).

Solution: $B = 7/15 \times 6,900 = ₹3,220$.

Trap: ₹3,680 — reporting A's share instead of B's.

Q63. In a school the ratio of boys to girls is 4 : 5. If 10 more boys are admitted and 10 girls leave, the ratio becomes 6 : 5. How many students were there originally?

A. 108 B. 81 C. 90 D. 99

Answer: D — 99

Trick: $(4x + 10)/(5x - 10) = 6/5 \rightarrow 20x + 50 = 30x - 60 \rightarrow x = 11 \rightarrow 9 \times 11 = 99$.

Solution: 99 students.

Trap: 90 — rounding x to 10 while solving the equation.

Q64. A's share to B's is 3 : 4 and B's to C's is 5 : 6. If the total of their shares is ₹1,180, find C's share.

A. ₹480 B. ₹500 C. ₹400 D. ₹300

Answer: A — ₹480

Trick: Link through B: $A : B : C = 15 : 20 : 24 = 59$ parts; unit = $1,180/59 = 20$.

Solution: $C = 24 \times 20 = ₹480$.

Trap: ₹400 — reporting B's share.

Q65. A and B invest ₹20,000 and ₹30,000. After 6 months A withdraws ₹5,000. Find the ratio of their profits at the end of the year.

A. 7 : 10 B. 7 : 12 C. 12 : 7 D. 2 : 3

Answer: B — 7 : 12

Trick: A: $20,000 \times 6 + 15,000 \times 6 = 2,10,000$; B: $30,000 \times 12 = 3,60,000 \rightarrow 7 : 12$.

Solution: A: $20,000 \times 6 + 15,000 \times 6 = 2,10,000$; B: $3,60,000 \rightarrow 7 : 12$.

Trap: 2 : 3 — using the initial capitals only and ignoring the withdrawal.

Q66. Two numbers are in the ratio 5 : 8. If 10 is subtracted from each, the ratio becomes 3 : 5. Find the numbers.

A. 75 and 120 B. 50 and 80 C. 40 and 64 D. 100 and 160

Answer: D — 100 and 160

Trick: $(5x - 10)/(8x - 10) = 3/5 \rightarrow 25x - 50 = 24x - 30 \rightarrow x = 20$.

Solution: Numbers are 100 and 160.

Trap: 50 and 80 — reading the original ratio numbers ($x = 10$) as the answer.

Q67. ₹1,710 is divided among A, B and C such that $A : B = 5 : 6$ and $B : C = 3 : 4$. Find C's share.

A. ₹450 B. ₹540 C. ₹630 D. ₹720

Answer: D — ₹720

Trick: $A : B : C = 15 : 18 : 24 = 57$ parts; unit = $1,710/57 = 30$.

Solution: $C = 24 \times 30 = ₹720$.

Trap: ₹540 — reporting B's share.

Q68. A and B invest ₹50,000 and ₹75,000 in a firm. For managing the firm A gets 10% of the profit, and the rest is divided in the investment ratio. If the annual profit is ₹24,000, what does A get in all?

A. ₹12,000 B. ₹12,960 C. ₹11,040 D. ₹8,640

Answer: C — ₹11,040

Trick: Peel the fee first: ₹2,400 to A; split the remaining ₹21,600 as 2 : 3 → A adds ₹8,640.

Solution: $A = 2,400 + 8,640 = ₹11,040$.

Trap: ₹8,640 — forgetting the management fee.

Q69. What is the duplicate ratio of 2 : 3?

A. 4 : 6 B. 4 : 9 C. 2 : 3 D. 8 : 27

Answer: B — 4 : 9

Trick: Duplicate = square each term: $2^2 : 3^2 = 4 : 9$.

Solution: $2^2 : 3^2 = 4 : 9$.

Trap: 4 : 6 — doubling each term instead of squaring.

Q70. A bag contains 50-paise, 25-paise and 10-paise coins in the ratio 5 : 9 : 4, amounting to ₹206 in all. How many 25-paise coins are there?

A. 360 B. 200 C. 160 D. 320

Answer: A — 360

Trick: Value per set: $5(0.5) + 9(0.25) + 4(0.1) = 5.15 \rightarrow x = 206/5.15 = 40$.

Solution: 25-paise coins = $9 \times 40 = 360$.

Trap: 200 — reporting the number of 50-paise coins.

Q71. Seats for Mathematics, Physics and Biology are in the ratio 5 : 7 : 8. They are increased by 40%, 50% and 75% respectively. What is the new ratio?

A. 7 : 10 : 14 B. 5 : 7 : 8 C. 2 : 3 : 4 D. 3 : 4 : 5

Answer: C — 2 : 3 : 4

Trick: New = old $\times (1 + \text{rise})$: 7 : 10.5 : 14 → double everything → 2 : 3 : 4.

Solution: $5 \times 1.4 : 7 \times 1.5 : 8 \times 1.75 = 7 : 10.5 : 14 = 2 : 3 : 4$.

Trap: 7 : 10 : 14 — rounding 10.5 down instead of clearing the half.

Q72. If $(a + b) : (b + c) : (c + a) = 6 : 7 : 8$ and $a + b + c = 14$, find the value of c.

A. 8 B. 4 C. 6 D. 3

Answer: C — 6

Trick: The three pair-sums add to $2(a+b+c)$: $21k = 28 \rightarrow k = 4/3$; $c = (7k + 8k - 6k)/2 = 6$.

Solution: $6k + 7k + 8k = 28 \rightarrow k = 4/3$; $c = (7k + 8k - 6k)/2 = 6$.

Trap: 4 — reporting a's value ($6k/2$) instead of c's.

Q73. A invests ₹3,000 for 12 months, B ₹6,000 for 9 months and C ₹9,000 for 6 months. Out of a profit of ₹8,600, what does B receive?

A. ₹2,150 B. ₹2,580 C. ₹3,010 D. ₹3,225

Answer: D — ₹3,225

Trick: Capital $\times$ time = 36 : 54 : 54 (thousand-months) = 2 : 3 : 3; unit = $8,600/8 = 1,075$.

Solution: $B = 3 \times 1,075 = ₹3,225$.

Trap: ₹2,150 — reporting A's share.

Q74. If $x : y = 3 : 2$, find $(2x + 3y) : (3x - y)$.

A. 7 : 12 B. 9 : 7 C. 12 : 5 D. 12 : 7

Answer: D — 12 : 7

Trick: Substitute $x = 3$, $y = 2$ and simplify: 12 : 7.

Solution: $2(3) + 3(2) = 12$; $3(3) - 2 = 7 \rightarrow 12 : 7$.

Trap: 7 : 12 — reading the ratio in reverse.

Q75. In a partition of ₹3,600, A's share equals half of B's and C's shares together, and B's share equals one-third of C's and A's shares together. Find A's share.

A. ₹1,200 B. ₹900 C. ₹1,680 D. ₹1,500

Answer: A — ₹1,200

Trick: $2A = B + C$ and $3B = A + C$ solve to $A : B : C = 4 : 3 : 5$ (12 parts of ₹300).

Solution: $A = 4 \times 300 = ₹1,200$.

Trap: ₹1,500 — reporting C's share.

Q76. The average of 8 numbers is 12. If each number is increased by 3, what is the new average?

A. 12 B. 24 C. 15 D. 9

Answer: C — 15

Trick: A constant added to every number shifts the average by the same amount.

Solution: New average = $12 + 3 = 15$.

Trap: 24 — multiplying the shift by 8 (3×8 added to 12).

Q77. The average of 10 numbers is 15. If one number 18 is removed, what is the new average?

A. 15 B. 14 C. $14\frac{2}{3}$ D. 13.8

Answer: C — $14\frac{2}{3}$

Trick: New sum = $150 - 18 = 132$ over the new count 9.

Solution: $132/9 = 14\frac{2}{3}$.

Trap: 15 — assuming the average is unchanged.

Q78. The average weight of 8 persons increases by 2.5 kg when a new person weighing 65 kg replaces one weighing 45 kg. How many persons are in the group?

A. 8 B. 10 C. 6 D. 12

Answer: A — 8

Trick: Total rise = swap difference $\div$ per-head rise = $20 \div 2.5 =$ number of members.

Solution: $n = 20/2.5 = 8$.

Trap: 10 — dividing 20 by 2 and dropping the decimal.

Q79. The average age of 30 students is 12 years. If the teacher's age is included, the average increases by 1 year. Find the teacher's age.

A. 42 years B. 43 years C. 33 years D. 40 years

Answer: B — 43 years

Trick: Teacher's age = new average + total headcount $\times$ rise = $13 + 31 \times 1$ — the +1 shows up once for the teacher and 31 times for the class.

Solution: $12 + 31 \times 1 = 43$.

Trap: 42 — multiplying the rise by 30 instead of 31.

Q80. The average age of A and B is 20 years. If C replaces A, the average becomes 19 years; if C replaces B, it becomes 21 years. Find A's age.

A. 22 years B. 19 years C. 20 years D. 18 years

Answer: A — 22 years

Trick: $A = [(A+B) + (A+C) - (B+C)]/2 = (40 + 42 - 38)/2$ — A appears in two of the three pair-sums.

Solution: $A = 22$ years.

Trap: 18 years — assigning A the +2 branch instead of the -2 one.

Q81. The present ages of A and B are in the ratio 4 : 5. Eight years from now the ratio will be 5 : 6. Find B's present age.

A. 36 years B. 40 years C. 45 years D. 32 years

Answer: B — 40 years

Trick: Equal absolute change: Δ units = $6 - 5 = 1 \rightarrow 8$ years = 1 old unit $\rightarrow A = 32, B = 40$.

Solution: $B = 40$ years.

Trap: 36 years — solving for the FUTURE ratio's unit (1 unit = 8 future years) and misapplying it.

Q82. The average of the first 20 natural numbers is:

A. 10.5 B. 20 C. 10 D. 11

Answer: A — 10.5

Trick: Average of $1..n = (n + 1)/2$.

Solution: $(20 + 1)/2 = 10.5$.

Trap: 10 — dropping the .5 and rounding down.

Q83. The average of 11 results is 50. If the average of the first 6 results is 49 and that of the last 6 is 52, find the 6th result.

A. 90 B. 100 C. 56 D. 50

Answer: C — 56

Trick: The 6th result lies in BOTH groups: $6(49) + 6(52) - 11(50) =$ the 6th result.

Solution: $294 + 312 - 550 = 56$.

Trap: 100 — adding the two group totals instead of subtracting the overall total.

Q84. The average of 5 numbers is 27. If one number is excluded, the average of the remaining numbers becomes 25. Find the excluded number.

A. 27 B. 25 C. 35 D. 30

Answer: C — 35

Trick: Excluded number = old total - new total = $135 - 100$.

Solution: $5 \times 27 - 4 \times 25 = 135 - 100 = 35$.

Trap: 30 — subtracting the averages ($27 - 25 = 2 \times 5 = 10$ added wrongly).

Q85. The average monthly income of a family of 4 members is ₹12,000. A new member joins and the average drops

to ₹11,500. What is the new member's income?

A. ₹11,000 B. ₹11,500 C. ₹9,500 D. ₹10,000

Answer: C — ₹9,500

Trick: New member = new total – old total.

Solution: ₹57,500 – ₹48,000 = ₹9,500.

Trap: ₹10,000 — multiplying the drop (₹500) by 5 instead of 4.

Q86. Five years ago, the average age of a family of 4 members was 25 years. A baby has been born in the meantime. What is the present average age of the family?

A. 20 years B. 24 years C. 25 years D. 30 years

Answer: B — 24 years

Trick: Old total $100 + 4 \times 5$ years of aging = 120; the baby adds 0 but raises the headcount to 5.

Solution: $120/5 = 24$ years.

Trap: 30 years — dividing the new total 120 by the old headcount 4 (forgetting the baby).

Q87. The average temperature of Monday, Tuesday and Wednesday was 37°C, and of Tuesday, Wednesday and Thursday was 34°C. If the temperature on Thursday was $\frac{4}{5}$ that of Monday, find Thursday's temperature.

A. 36°C B. 45°C C. 40°C D. 30°C

Answer: A — 36°C

Trick: Subtract the two 3-day sums: Monday – Thursday = $37 \times 3 - 34 \times 3 = 9$. With $Th = (\frac{4}{5})M$, $M/5 = 9 \rightarrow M = 45$.

Solution: Thursday = $\frac{4}{5} \times 45 = 36^\circ\text{C}$.

Trap: 45°C — reporting Monday's temperature, which the question never asked for.

Q88. A cricketer's average in 10 innings is 38.9. How many runs must he score in the 11th innings to raise his average to 40?

A. 45 runs B. 61 runs C. 51 runs D. 40 runs

Answer: C — 51 runs

Trick: Required runs = new total – old total.

Solution: $440 - 389 = 51$ runs.

Trap: 45 runs — taking $40 +$ the deficit $(1.1) \times 10 = 51$ vs a rough $40 + 5$ guess.

Q89. The average salary of workers in an office is ₹6,000. The average salary of 15 technicians is ₹8,000 and the average of the rest is ₹5,000. Find the total number of workers.

A. 50 B. 30 C. 60 D. 45

Answer: D — 45

Trick: Each technician pulls the average UP ₹2,000; each other worker pulls it DOWN ₹1,000 — 15 techs need 30 others: total 45.

Solution: $15 + 30 = 45$ workers.

Trap: 30 — counting only the non-technicians and forgetting the 15 technicians.

Q90. The average age of a husband and wife at the time of their marriage was 23 years. After 5 years they have a 3-year-old child. What is the present average age of the family?

A. 19.67 years B. 23 years C. 21 years D. 17 years

Answer: A — 19.67 years

Trick: Couple total at marriage = 46; add $5 + 5$ years of aging and the 3-year-old child $\rightarrow 59$ over 3 members.

Solution: $59/3 \approx 19.67$ years.

Trap: 23 years — reporting the couple's average at marriage and ignoring both aging and the child.

Q91. The average age of 8 members of a team is 30 years. A new member joins, replacing one who is 34 years old, and the average becomes 31 years. Find the new member's age.

A. 38 years B. 35 years C. 40 years D. 42 years

Answer: D — 42 years

Trick: New total – old total + outgoing member = incoming member.

Solution: $248 - 240 + 34 = 42$ years.

Trap: 38 years — adding the outgoing member's age instead of subtracting it ($248 - 240 - 34$).

Q92. A batsman makes a score of 87 in the 17th innings and thus increases his average by 3. Find his average after the 17th innings.

A. 36 B. 42 C. 40 D. 39

Answer: D — 39

Trick: Let the new average be x : $16(x - 3) + 87 = 17x \rightarrow x = 87 - 48 = 39$.

Solution: New average = 39.

Trap: 42 — solving for the OLD average ($x + 3$) and reporting it.

Q93. The average weight of A, B and C is 45 kg. The average of A and B is 40 kg and that of B and C is 43 kg. Find B's weight.

A. 38 kg B. 41 kg C. 31 kg D. 43 kg

Answer: C — 31 kg

Trick: $B = (A+B) + (B+C) - (A+B+C)$: $80 + 86 - 135 = 31$ — B is the only name in both pair-sums.

Solution: $B = 80 + 86 - 135 = 31$ kg.

Trap: 41 kg — taking the plain average of the two pair-sums instead of subtracting the triple total.

Q94. A man's average expenditure for the first 4 months is ₹1,800, for the next 5 months ₹2,100, and he saves ₹1,308 in the last 3 months. Find his average monthly income.

A. ₹1,600 B. ₹1,584 C. ₹2,100 D. ₹1,500

Answer: B — ₹1,584

Trick: Income = spending + saving; add the three blocks ($7,200 + 10,500 + 1,308$) and divide by 12.

Solution: $19,008/12 = ₹1,584$.

Trap: ₹1,500 — dropping the ₹1,308 saving and slipping the spend-sum to ₹18,000 ($7,200 + 10,500 = 17,700$): $18,000 \div 12$.

Q95. The average age of a family of 5 members is 24 years. If the youngest member is 8 years old, what was the average age of the family at the birth of the youngest member?

A. 16 years B. 20 years C. 22.4 years D. 21.6 years

Answer: A — 16 years

Trick: 8 years ago all FOUR other members were 8 years younger and the newborn counts as 0: $(120 - 8 - 4 \times 8)/5 = 80/5 = 16$.

Solution: Total now 120; minus the youngest (8) minus 8×4 for the other four $\rightarrow 80$; $\div 5$ members (newborn included) = 16 years.

Trap: 22.4 — subtracting only the youngest's own age; 20 — dividing by 4 (dropping the newborn from the count).

Q96. The average age of A, B, C is 26 years. If A is 28 years old and C is 30 years old, find B's age.

A. 26 years B. 24 years C. 20 years D. 22 years

Answer: C — 20 years

Trick: $B = 3 \times \text{average} - A - C = 78 - 58$.

Solution: $3 \times 26 - 28 - 30 = 20$.

Trap: 24 years — assuming the spread is symmetric around 26.

Q97. The average of 25 results is 20. The average of the first 12 results is 18 and that of the last 14 is 21. Find the 12th result.

A. 10 B. 20 C. 12 D. -10

Answer: A — 10

Trick: The 12th result sits in BOTH groups: $12(18) + 14(21) - 25(20) =$ the 12th result.

Solution: $216 + 294 - 500 = 10$.

Trap: -10 — subtracting the group sums from the total (reversing the overlap formula).

Q98. The average age of 36 students in a group is 14 years. When the teacher's age is included, the average increases by one. What is the teacher's age in years?

A. 51 years B. 50 years C. 53 years D. 49 years

Answer: A — 51 years

Trick: Teacher = new average + (new count) $\times$ rise = $15 + 37$.

Solution: $15 + 37 \times 1 = 51$.

Trap: 50 years — multiplying the rise by 36 instead of 37.

Q99. The average height of 30 boys was calculated as 150 cm. Later it was discovered that one boy's height was recorded as 165 cm instead of 145 cm. Find the correct average.

A. $149\frac{1}{3}$ cm B. 148 cm C. 150 cm D. 149 cm

Answer: A — $149\frac{1}{3}$ cm

Trick: Correct total = wrong total - (recorded - actual); correction = $20/30$ down.

Solution: $150 - 20/30 = 149\frac{1}{3}$.

Trap: 150 cm — assuming a single wrong entry can't shift the average.

Q100. The average age of 5 members of a family is 21 years. The youngest is 9 years old. What was the average age of the family at the birth of the youngest member?

A. 15 years B. 19.2 years C. 12 years D. 18.2 years

Answer: C — 12 years

Trick: 9 years ago all FOUR other members were 9 years younger and the newborn counts as 0: $(105 - 9 - 4 \times 9)/5 =$

$$60/5 = 12.$$

Solution: Total now 105; minus the youngest (9) minus 9×4 for the other four $\rightarrow 60$; $\div 5$ members = 12 years.

Trap: 19.2 — subtracting only the youngest's own age; 15 — dividing by 4 (dropping the newborn from the count).

Q101. In what ratio must rice at ₹9.30 per kg be mixed with rice at ₹10.80 per kg so that the mixture is worth ₹10 per kg?

A. 3 : 4 B. 8 : 7 C. 7 : 8 D. 9 : 7

Answer: B — 8 : 7

Trick: Alligation: cheaper : dearer = $(10.80 - 10) : (10 - 9.30) = 0.80 : 0.70 = 8 : 7$.

Solution: $(10.80 - 10) : (10 - 9.30) = 0.8 : 0.7 = 8 : 7$.

Trap: 7 : 8 — putting the dearer variety first.

Q102. A vessel contains 60 litres of milk and water in the ratio 7 : 3. How much water must be added to make the ratio 7 : 5?

A. 6 L B. 12 L C. 10 L D. 18 L

Answer: B — 12 L

Trick: Milk is fixed at 42 L: water must reach $5/7 \times 42 = 30$ L $\rightarrow$ add 12 L.

Solution: Water needed $30 - 18 = 12$ L.

Trap: 10 L — setting $7/12$ of the NEW total equal to 42 L and solving for the wrong quantity.

Q103. A 20-litre mixture contains milk and water in the ratio 3 : 1. How much water must be added to make the ratio 3 : 2?

A. 5 L B. 4 L C. 6 L D. 10 L

Answer: A — 5 L

Trick: Milk 15 L is fixed: water must become $2/3 \times 15 = 10$ L $\rightarrow$ add $10 - 5 = 5$ L.

Solution: Water needed $10 - 5 = 5$ L.

Trap: 4 L — treating one ratio unit (5 L) as the amount to add.

Q104. In what ratio should water be mixed with milk costing ₹60 per litre so that the mixture costs ₹40 per litre?

A. 1 : 1 B. 2 : 3 C. 1 : 2 D. 2 : 1

Answer: C — 1 : 2

Trick: Water at ₹0: water : milk = $(60 - 40) : (40 - 0) = 20 : 40 = 1 : 2$.

Solution: $(60 - 40) : (40 - 0) = 1 : 2$.

Trap: 2 : 1 — reversing milk and water in the alligation cross.

Q105. A milkman has 40 L of milk worth ₹36 per litre. How much water (free) must be added so that the mixture sells at ₹32 per litre with no profit no loss?

A. 8 L B. 4 L C. 6 L D. 5 L

Answer: D — 5 L

Trick: Total value ₹1,440 is fixed: new volume = $1,440/32 = 45$ L $\rightarrow$ add 5 L.

Solution: New volume $1,440/32 = 45$ L $\rightarrow 45 - 40 = 5$ L.

Trap: 4 L — adding 10% of the volume instead of solving the value equation.

Q106. Two vessels contain milk and water in the ratios 4 : 5 and 5 : 1. In what ratio should their contents be mixed to get a new mixture with milk and water in the ratio 5 : 4?

A. 5 : 2 B. 1 : 2 C. 2 : 5 D. 5 : 6

Answer: A — 5 : 2

Trick: Alligation on milk fractions: vessel1 : vessel2 = $(5/6 - 5/9) : (5/9 - 4/9) = 5/18 : 2/18 = 5 : 2$.

Solution: $(5/6 - 5/9) : (5/9 - 4/9) = 1/18 : 2/18 \rightarrow$ vessel1 : vessel2 = 5 : 2.

Trap: 2 : 5 — reading the alligation cross backwards.

Q107. A vessel contains 56 litres of milk. 7 litres are drawn out and replaced with water. This operation is done twice more. Find the ratio of milk to water in the final mixture.

A. 7 : 9 B. 1 : 1 C. 3 : 4 D. 343 : 169

Answer: D — 343 : 169

Trick: Milk left = $56 \times (1 - 7/56)^3 = 56 \times (7/8)^3 = 56 \times 343/512 = 37.44$ L $\rightarrow$ fraction $343/512 \rightarrow$ ratio 343 : 169.

Solution: Milk : water = 343 : $(512 - 343) = 343 : 169$.

Trap: 7 : 9 — assuming 21 L of milk is removed in total (simple subtraction).

Q108. A container has 80 L of milk. 8 L is replaced with water, and the process is repeated once more. How much milk remains?

A. 64 L B. 66 L C. 72 L D. 64.8 L

Answer: D — 64.8 L

Trick: Milk = $80 \times (1 - 8/80)^2 = 80 \times 0.81$.

Solution: $80 \times 0.9 \times 0.9 = 64.8$ L.

Trap: 64 L — subtracting $8 + 8 = 16$ L outright.

Q109. y varies directly as x . If $y = 45$ when $x = 9$, find y when $x = 15$.

A. 60 B. 81 C. 90 D. 75

Answer: D — 75

Trick: Direct: $y/x = k = 5 \rightarrow y = 5 \times 15$.

Solution: $k = 45/9 = 5 \rightarrow y = 5 \times 15 = 75$.

Trap: 60 — computing $45 + (15 - 9) \times k$ where $k = 3$? No — $45 + 6 \times 5 = 75$; the trap is a DIFFERENT k read as 3.

Q110. x varies inversely as y . If $x = 12$ when $y = 5$, find x when $y = 15$.

A. 20 B. 8 C. 4 D. 36

Answer: C — 4

Trick: Inverse: $xy = k = 60 \rightarrow x = 60/15$.

Solution: $k = 12 \times 5 = 60 \rightarrow x = 60/15 = 4$.

Trap: 36 — treating the relation as direct (12×3).

Q111. If x varies as the square of y and $x = 16$ when $y = 2$, find x when $y = 5$.

A. 100 B. 80 C. 64 D. 40

Answer: A — 100

Trick: $x/y^2 = k = 4 \rightarrow x = 4 \times 5^2$.

Solution: $k = 16/2^2 = 4 \rightarrow x = 4 \times 5^2 = 100$.

Trap: 40 — using direct (linear) variation $k = 8 \rightarrow x = 40$.

Q112. If 15 workers can finish a job in 12 days, how many days will 20 workers take?

A. 9 days B. 16 days C. 10 days D. 8 days

Answer: A — 9 days

Trick: Inverse variation: workers $\times$ days = 180 $\rightarrow$ days = 180/20.

Solution: $180/20 = 9$ days.

Trap: 16 days — reading the ratio as direct ($20/15 \times 12$).

Q113. The cost of 8 kg of rice is ₹240. What is the cost of 13 kg?

A. ₹390 B. ₹360 C. ₹400 D. ₹420

Answer: A — ₹390

Trick: Direct variation: cost/kg = 30 $\rightarrow 13 \times 30$.

Solution: $240/8 \times 13 = ₹390$.

Trap: ₹360 — taking a proportional guess of 12 kg instead of 13.

Q114. In what ratio must water be mixed with milk costing ₹60 per litre to obtain a mixture worth ₹42 per litre?

A. 2 : 5 B. 3 : 7 C. 7 : 3 D. 6 : 7

Answer: B — 3 : 7

Trick: Water : milk = $(60 - 42) : (42 - 0) = 18 : 42 = 3 : 7$.

Solution: $(60 - 42) : (42 - 0) = 18 : 42 = 3 : 7$.

Trap: 7 : 3 — reversing milk and water in the alligation cross.

Q115. A vessel contains 63 litres of a milk-water mixture with milk and water in the ratio 5 : 4. How much water must be added to make the ratio 5 : 6?

A. 14 L B. 7 L C. 9 L D. 21 L

Answer: A — 14 L

Trick: Milk 35 L is fixed: water must become $6/5 \times 35 = 42$ L $\rightarrow$ add $42 - 28 = 14$ L.

Solution: Water must reach $42 - 28 = 14$ L.

Trap: 7 L — treating one ratio unit (7 L) as the addition.

Q116. In what ratio must tea at ₹320 per kg be mixed with tea at ₹250 per kg so that the mixture costs ₹290 per kg?

A. 5 : 7 B. 3 : 4 C. 2 : 3 D. 4 : 3

Answer: D — 4 : 3

Trick: ₹320-tea : ₹250-tea = $(290 - 250) : (320 - 290) = 40 : 30 = 4 : 3$ — each variety takes the gap on the OPPOSITE side.

Solution: 4 : 3 (290 sits closer to ₹320, so more of the dearer tea).

Trap: 3 : 4 — swapping the two varieties.

Q117. A dishonest milkman mixes water with milk and sells the mixture at cost price, gaining 20%. What fraction of the mixture is water?

A. $1/3$ B. $1/6$ C. $1/5$ D. $1/4$

Answer: B — $1/6$

Trick: Selling at cost with $g\%$ gain $\rightarrow$ water fraction = $g/(100 + g) = 20/120 = 1/6$.

Solution: water fraction = $20/120 = 1/6$.

Trap: $1/5$ — equating the gain % ($20\% = 1/5$) directly with the water fraction.

Q118. If 12 men or 18 women can do a piece of work in 20 days, in how many days can 16 men and 12 women do it?

A. 10 days B. 12 days C. 18 days D. 15 days

Answer: A — 10 days

Trick: 1 man = 1.5 women $\rightarrow 16\text{ M} + 12\text{ W} = 16 + 8 = 24$ men-equivalents $\rightarrow 240/24 = 10$ days.

Solution: $12\text{M} = 18\text{W} \rightarrow 1\text{M} = 1.5\text{W}$; $16\text{M} + 12\text{W} = 36\text{W}$ -eq; work = $18 \times 20 = 360$ W-days $\rightarrow 360/36 = 10$ days.

Trap: 12 days — dividing by the 20-man count and ignoring the women.

Q119. A mixture of 45 L contains milk and water in the ratio 4 : 1. How much water must be added to make the ratio 3 : 2?

A. 15 L B. 18 L C. 9 L D. 12 L

Answer: A — 15 L

Trick: Milk 36 L is fixed: water must reach $2/3 \times 36 = 24$ L $\rightarrow$ add $24 - 9 = 15$ L.

Solution: Water must reach $24 - 9 = 15$ L.

Trap: 9 L — the existing water amount mistaken for the addition.

Q120. The speed of a train varies inversely as the time taken for a fixed journey. At 60 km/h the journey takes 5 hours. How long does it take at 50 km/h?

A. 5.5 hours B. 4 hours C. 7.2 hours D. 6 hours

Answer: D — 6 hours

Trick: Inverse: speed $\times$ time = 300 km fixed $\rightarrow$ time = $300/50$.

Solution: $300\text{ km} = 60 \times 5 \rightarrow t = 300/50 = 6$ hours.

Trap: 4 hours — reading the variation as direct.